

The Dark Sector as a Classification Error

A Boundary Analysis of Dark Matter and Dark Energy

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January 4, 2026

Abstract

Dark matter and dark energy are usually introduced through empirically successful metric cosmological models, especially Λ CDM. The usual inference is then strengthened into an ontological claim: because additional gravitating or accelerating terms are indispensable to the model, its sector terms must be invariantly extracted as primitive constituents of the universe. This paper argues that the strengthening is not admissible. The target is narrow. We do not deny galactic dynamics, gravitational lensing, the cosmic microwave background, baryon acoustic structure, supernova acceleration, or the predictive success of Λ CDM. We ask only whether the role terms “dark matter” and “dark energy”, as they function in metric surrogate inference, achieve representation-independent extraction sufficient for primitive ontological status.

The answer is negative. A dark-sector term is introduced as a stabilizing residual inside a non-invariant decomposition of witness constraints into metric geometry, known stress-energy, auxiliary source terms, boundary data, calibration records, and error discipline. Over the restricted observational regimes motivating the corresponding dark-sector inference, and for fixed comparison targets, admissible redescription can preserve the relevant witness content while redistributing the stabilizing load among these coordinates: an added stress-energy component, a modified gravitational law, an effective curvature term, an emergent variable, a calibration or transport correction, or a changed boundary ledger. This is not a claim of global empirical equivalence among complete cosmological theories. It is a local non-extraction result: a term whose identity is fixed only by this load-bearing role is not standing-preserving. It is a surrogate-dependent stabilization structure, not an entity-class thereby invariantly extracted.

The resulting conclusion is conditional but sharp. If a future particle, field, or invariant structure is extracted by a standing-preserving witness protocol, that object may receive its own ontological assessment. The presently inferred dark-sector structure does not receive that status from current metric-surrogate indispensability. Dark-sector terms have predictive necessity inside successful representations; they do not thereby achieve invariant primitive ontological extraction. Treating the presently inferred dark-sector structure as already invariantly extracted primitive structure is a classification error: it mistakes a surrogate stabilizer for an invariantly extracted entity-class. The result is stronger than ordinary underdetermination because it identifies a structural failure of invariant extraction, not merely a plurality of empirically adequate stories.

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1 Jurisdiction, target, and non-claims

This paper is a Category A boundary analysis. It does not propose a replacement cosmology, fit a new data set, alter the Einstein equations, adjudicate between Λ CDM and modified-gravity programs, or deny the empirical phenomena for which the dark sector was introduced. The paper's sole question is classificatory:

Given the role played by dark matter and dark energy in metric cosmological inference, are these role terms licensed for primitive ontological extraction?

The answer defended here is no. The conclusion is deliberately not the stronger claim that no undiscovered particle exists, that no additional field can ever be detected, or that all dark-sector model building is scientifically empty. Particle searches, precision cosmology, modified gravity, and effective-field-theory work remain legitimate empirical programs. The present argument does not deny the possibility of ontology in cosmology; it establishes a constraint on when ontological extraction is admissible from surrogate-dependent decompositions. The present argument targets primitive extraction from surrogate-dependent decomposition, not the operational utility of effective cosmological entities within constrained predictive regimes. The present analysis does not require adoption of any broader metaphysical system. Only the minimal distinction between predictive

surrogate structure and invariant ontological extraction is assumed. The point is that predictive indispensability inside a surrogate model is not sufficient for invariant ontological extraction.

Nor is the result merely the familiar observation that several empirically adequate theories may coexist. Ordinary underdetermination is usually epistemic: more than one theory may fit a given body of evidence. The boundary result here is structural: when a role term is defined only by a decomposition-dependent residual and no standing-preserving extraction map fixes its identity across admissible redescrptions, invariant extraction fails in the fixed scope, not merely optional.

This distinction matters because the dark sector has two different uses. In the modest use, “dark matter” and “dark energy” name missing terms in a successful compression of observations. In the stronger use, they name constituents of reality. The first use is methodologically sound. The second use requires an additional invariant extraction that the role terms themselves do not supply.

Remark 1 (Meaning of classification error). *A classification error, in this paper, is not an empirical mistake. It is a type mistake in the promotion of a term. A term that belongs to a representational layer is classified as though it had already achieved standing-preserving extraction. The target is therefore the promotion rule, not the observations.*

2 Orthodox cosmological setting

The standard cosmological setting is familiar. Galaxy rotation curves, cluster dynamics, weak and strong lensing, large-scale structure, the acoustic structure of the cosmic microwave background, and structure-formation constraints support the introduction of an additional non-luminous gravitating component when the data are interpreted inside general-relativistic metric cosmology [1–5, 11]. Type Ia supernovae, CMB-plus-BAO constraints, and other distance-redshift measurements support a late-time acceleration term, commonly represented in the minimal model by a cosmological constant [6, 7, 9, 11–13]. The resulting Λ CDM model is empirically powerful and has become the baseline compression of much cosmological data.

The boundary claim here is compatible with that success. What it rejects is a common extra step:

The model needs a role term; therefore the role term has been invariantly extracted as a primitive constituent.

The inference is not forced. In standard form one writes, schematically,

$$G_{\mu\nu}[g] + \Lambda g_{\mu\nu} = 8\pi G \left(T_{\mu\nu}^b + T_{\mu\nu}^\gamma + T_{\mu\nu}^\nu + T_{\mu\nu}^{\text{DM}} \right), \quad (1)$$

where baryons, photons, neutrinos, a cold-dark-matter component, and a cosmological constant are assigned distinct coordinates in the model. At the homogeneous level one often writes

$$H^2(a) = H_0^2 \left[\Omega_r a^{-4} + \Omega_m a^{-3} + \Omega_k a^{-2} + \Omega_\Lambda \right] \quad (2)$$

for the minimal spatially curved background parameterization, with additional structure in perturbations, nuisance parameters, likelihoods, and calibration records.

Equations (1) and (2) are not in dispute. The issue is what follows from them. A metric surrogate contains a decomposition of the world into geometric variables, source variables, background

assumptions, perturbative variables, record-selection choices, and error structures. A coordinate in that decomposition does not automatically achieve invariant ontological extraction from the empirical success of the whole surrogate.

3 Witnesses, surrogates, and standing

We need a small amount of terminology. It is chosen to be compatible with ordinary cosmological inference while preserving the AASC boundary distinction between admissible witness content and presentation-dependent stabilization structure.

Definition 1 (Witness constraints). *A witness constraint is any observational, experimental, or record-level constraint that the cosmological account must preserve: lensing maps, rotation curves, cluster velocities, CMB spectra, BAO distances, supernova distance moduli, abundance constraints, calibration records, likelihood-building records, and declared systematic-error envelopes. The set of witness constraints under discussion is denoted by $\mathcal{W}$.*

Definition 2 (Metric surrogate). *A metric cosmological surrogate is a tuple*

$$\mathcal{S} = (\mathcal{M}, g_{\mu\nu}, \Theta, \mathbf{P}, \mathbf{I}, \mathbf{R}, \mathbf{E}), \quad (3)$$

where $\mathcal{M}$ is a model family, $g_{\mu\nu}$ is the metric structure used by that family, Θ is its parameter space, $\mathbf{P}$ is the prediction map from model-plus-parameters to witnesses, $\mathbf{I}$ is the inference map from witnesses to parameters or posteriors, $\mathbf{R}$ is the record and calibration ledger, and $\mathbf{E}$ is the declared error discipline.

Definition 3 (Admissible redescription). *For the purposes of the present analysis, an admissible redescription is a transformation of the surrogate presentation that preserves the observational witness structure motivating the relevant cosmological inference and the declared comparison target. It may change coordinates, move terms between geometry and effective source, replace a source residual by a modified law, alter an auxiliary-field representation, or change a representational split, provided the relevant witness content is preserved within the fixed scope. This is a restricted same-witness condition, not a claim that complete cosmological theories are globally or exactly empirically equivalent.*

Definition 4 (Standing). *A term has standing only if its identity is invariant under admissible redescription and is recoverable by a standing-preserving witness protocol. In standard scientific language, this means representation-independent reconstructability: invariance of ontological extraction across admissible representational schemes preserving the relevant observational constraints. A term whose identity varies while the relevant witnesses are preserved has no standing in the fixed scope. It may be useful, indispensable, or computationally efficient; it is not thereby extracted as a primitive entity-class.*

Definition 5 (Standing-preserving extraction). *A standing-preserving extraction protocol is a reconstruction map from admissible witness constraints to an entity-class whose identity is invariant under every admissible redescription preserving the fixed witness set and comparison target. Formally, an extraction map*

$$E : \mathcal{W} \longrightarrow \mathcal{C} \quad (4)$$

is standing-preserving in the fixed scope only if, for every admissible redescription $R : \mathcal{S} \rightarrow \mathcal{S}'$ with $R(\mathcal{W}) = \mathcal{W}$, there exists an identity-preserving comparison ι_R such that

$$\iota_R(E_{\mathcal{S}}(\mathcal{W})) = E_{\mathcal{S}'}(\mathcal{W}). \quad (5)$$

If the recovered class changes, disappears, or is reclassified merely by moving between geometry, source, law, auxiliary field, boundary data, or inference ledger, extraction has failed. A residual coordinate may then remain predictively useful without being ontologically extracted.

For this paper, admissibility is local to observational witness preservation; no global equivalence relation over all possible cosmological theories is being assumed.

This definition does not trivialize scientific inference. Electrons, protons, photons, and many fields are not merely whatever makes a given equation close. They participate in multiple cross-stabilizing witness protocols with invariant identity across redescriptions. The question is whether the presently inferred dark-sector structure, as a sector-level construction, has this property.

4 Metric representation and non-invariant decompositions

The key boundary point is not that metric methods are invalid. It is that metric description is a licensed representation of physical structure, not a primitive generator of all ontology. In the AASC corpus this is the distinction between standing-bearing tensor content and skin [21–23]: a presentation layer may carry essential calculational structure while still lacking authority to generate primitive entities by itself.

The present claim is therefore not that metric structure lacks operational utility, empirical discipline, or physical significance. General-relativistic metric modeling remains the orthodox language in which the relevant data are compressed. The narrower claim is that metric representation alone is insufficient to confer ontological standing upon entities introduced solely to stabilize metric-surrogate representation.

Metric cosmology is a disciplined representation of observations through a spacetime metric, stress-energy assignments, background symmetries, perturbative expansions, and likelihood construction. Within that representation, source terms and curvature terms are not merely measured; they are assigned to coordinates in a decomposition. A familiar example is the cosmological constant:

$$G_{\mu\nu}[g] + \Lambda g_{\mu\nu} = 8\pi G T_{\mu\nu} \quad \Longleftrightarrow \quad G_{\mu\nu}[g] = 8\pi G \left(T_{\mu\nu} + T_{\mu\nu}^{\Lambda} \right), \quad (6)$$

with

$$T_{\mu\nu}^{\Lambda} = -\frac{\Lambda}{8\pi G} g_{\mu\nu}. \quad (7)$$

The same formal term can be written as geometry or as effective stress-energy. This does not make the two presentations empirically useless; it shows that ontological classification is not fixed by the side of the equation on which the term is written.

A similar point holds more generally. Modified gravity theories, scalar-tensor theories, effective-fluid descriptions, and emergent-gravity proposals often transfer representational load between law, source, auxiliary field, boundary condition, and effective stress-energy [14–18]. The transfer may or may not succeed empirically in a given domain. But whenever a fixed witness subdomain and

comparison target are preserved while the alleged entity changes role or disappears, the role term cannot achieve primitive ontological extraction without an additional standing-preserving extraction.

5 The classification-error theorem

The following theorem states the boundary in general form.

Definition 6 (Role-defined stabilizer). *Let $\mathcal{S}$ be a surrogate for witness constraints $\mathcal{W}$. A term $x \in \Theta$ or component X of $\mathcal{S}$ is a role-defined stabilizer when its defining condition has the form*

$$P(\mathcal{S}, X) \approx \mathcal{W} \quad (8)$$

with X introduced as whatever additional structure is needed to stabilize the surrogate's predictions, without an independent standing-preserving witness protocol for X itself.

Lemma 1 (Residual dependence). *A role-defined stabilizer is decomposition-relative. Its value and interpretation depend on what the surrogate has already assigned to known sources, geometry, boundary conditions, calibration records, prediction maps, and error discipline.*

Proof. By definition, X is introduced as the residual needed after all other coordinates in $\mathcal{S}$ have been fixed. Change the fixed decomposition while preserving the same witness target, and the residual changes with it. Therefore the residual is relative to the decomposition. $\square$

Theorem 1 (Classification-error theorem). *Let X be a dark-sector role term in a metric surrogate $\mathcal{S}$. Suppose:*

- (i) X is introduced as a role-defined stabilizer for witness constraints $\mathcal{W}$;*
- (ii) there exists at least one admissible redescription $R \in \mathcal{R}$ that preserves $\mathcal{W}$ while changing, absorbing, or relocating the role played by X ;*
- (iii) no standing-preserving witness protocol extracts X with invariant identity across the admitted redescriptions.*

Then X is not invariantly extracted in the fixed scope. Treating X as an already extracted primitive entity-class is a classification error.

Proof. Assume for contradiction that X has standing. Then the identity of X must be invariant under admissible redescription and must not depend on a merely presentational decomposition. By (ii), however, an admissible redescription preserves the witness constraints while changing, absorbing, or relocating the role played by X . By (iii), there is no independent witness protocol that restores invariant identity for X across these redescriptions. Therefore either the standing of X depends on the presentation, contradicting standing invariance, or an additional discriminator is imported beyond the fixed witness scope, changing the subject. Both options are inadmissible for same-scope primitive extraction. Hence X fails standing-preserving extraction. $\square$

Corollary 1 (Predictive indispensability is insufficient). *A term may be indispensable to a chosen surrogate and still fail invariant ontological extraction. Indispensability establishes role necessity within the surrogate; it does not establish standing-preserving extraction under admissible redescription.*

6 Toy model: weak-field load transfer without invariant extraction

The following toy model gives a weak-field surrogate example. It is deliberately modest. It is not proposed as a viable rival cosmology, does not assert global equivalence between dark-matter and modified-gravity programs, and does not deny the empirical strength of Λ CDM. Its function is narrower: to display, in orthodox weak-field notation, why reproduction of a fixed witness profile does not by itself extract a dark-sector entity-class.

Let the relevant witness profile be a finite set of radial acceleration records,

$$\mathcal{W}_a = \{(r_i, a_i, M_b(< r_i), \epsilon_i)\}_{i=1}^N, \quad (9)$$

where a_i is the inferred radial acceleration at radius r_i , $M_b(< r_i)$ is the baryonic mass ledger inside that radius, and ϵ_i records the observational and modeling tolerance. The source-side weak-field surrogate introduces an additional density coordinate by

$$\nabla^2 \Phi_D = 4\pi G(\rho_b + \rho_D), \quad (10)$$

$$a_D(r_i) = -\partial_r \Phi_D(r_i), \quad |a_D(r_i) - a_i| \leq \epsilon_i. \quad (11)$$

In this presentation the compensating structure is the residual

$$\rho_D^S = \frac{1}{4\pi G} \nabla^2 \Phi_D - \rho_b. \quad (12)$$

The law-side surrogate instead leaves the source ledger baryonic and shifts compensating load into the response operator:

$$\nabla \cdot \left[\mu \left(\frac{|\nabla \Phi_F|}{a_0} \right) \nabla \Phi_F \right] = 4\pi G \rho_b, \quad (13)$$

$$a_F(r_i) = -\partial_r \Phi_F(r_i), \quad |a_F(r_i) - a_i| \leq \epsilon_i. \quad (14)$$

Equation (13) is used only as a schematic modified-response presentation. No claim is made that a particular μ , a particular a_0 , or any MOND-like completion reproduces every dark-matter witness. The point is the formal contrast: in (10) the mismatch is source-side; in (13) the mismatch is law-side.

Proposition 1 (Weak-field load-transfer normal form). *Fix the witness profile $\mathcal{W}_a$ and a declared comparison target consisting only of the recorded accelerations within the stated tolerances. Suppose two local surrogates $\mathcal{S}_D$ and $\mathcal{S}_F$ satisfy (11) and (14) on that support. Then $\mathcal{S}_D$ and $\mathcal{S}_F$ preserve the same declared witness profile while assigning the compensating load to different coordinates of the surrogate. Unless an additional witness protocol identifies ρ_D independently of the source-side residual role, the residual density coordinate ρ_D^S is not invariantly extracted from $\mathcal{W}_a$ alone.*

Proof. The witness profile fixes only the recorded accelerations, baryonic ledger, and tolerances in the declared scope. In $\mathcal{S}_D$, the departure between baryonic Newtonian prediction and a_i is assigned to ρ_D^S . In $\mathcal{S}_F$, the same declared acceleration records are absorbed into the response operator while no separate ρ_D -coordinate is extracted. Therefore the transformation from source-side compensation to law-side compensation preserves $\mathcal{W}_a$ but does not preserve the identity of the alleged dark-sector entity-class. A standing-preserving extraction would require an identity comparison ι_R carrying the

extracted class from $\mathcal{S}_D$ to the extracted class in $\mathcal{S}_F$. On the declared witness support there is no such class in $\mathcal{S}_F$. The missing discriminator must come from additional witnesses, a bridge theorem, or an independently stabilized particle/field protocol. Without that extra structure, the residual coordinate is a surrogate-dependent stabilization coordinate rather than an invariantly extracted primitive object. $\square$

This example is intentionally local. It does not show that every source-side dark-matter model has an equally successful law-side counterpart, and it does not weaken the empirical case for nonbaryonic gravitating structure inside GR-like model families. It shows only the inference boundary needed here: a residual that is defined as whatever makes a chosen weak-field source equation reproduce the witness profile has not, for that reason alone, been extracted as the same entity across witness-preserving surrogate presentations. The same normal form reappears in relativistic cosmology when compensating load is placed on the source side, the geometric side, an auxiliary-field ledger, a boundary/transport ledger, or an inference/calibration ledger.

7 Dark matter as residual gravitating role

Dark matter is often described as non-luminous matter inferred through gravitational effects. That description has a modest operational reading and a stronger ontological reading. The modest reading is unobjectionable: within a metric gravitational surrogate, observed motions and lensing require an additional gravitating contribution beyond luminous baryons and standard radiation. The stronger reading says that the residual contribution has already been invariantly extracted as a primitive sector.

In a Newtonian limit, the issue can be made explicit. Suppose a potential Φ is inferred from orbital or lensing constraints and the baryonic density ρ_b is estimated from luminous matter. The residual density assigned to dark matter has the schematic form

$$\rho_{\text{DM}}^{\mathcal{S}} = \frac{1}{4\pi G} \nabla^2 \Phi - \rho_b - \rho_{\text{other}}. \quad (15)$$

This is a residual relative to Poisson/GR dynamics, baryonic mass modeling, distance estimates, stellar mass-to-light ratios, gas modeling, lensing reconstruction, boundary assumptions, and error discipline. In relativistic cosmology, the same point appears in the decomposition of the stress-energy side of the Einstein equations and in the perturbation/transfer-function ledger used by CMB and structure inference.

The residual can be extremely stable inside a model family. That stability is what makes cold dark matter powerful. But residual stability inside a chosen decomposition is not the same as standing under admissible redescription. Consider the following table.

Witness domain		Metric-surrogate role	Possible load transfer		Standing result
Galaxy curves	rotation	Additional halo density stabilizes orbital velocities	Modified law, revision, auxiliary field	acceleration modeling environment-relation,	Role not invariant

Cluster dynamics	Missing gravitating mass stabilizes virial/lensing estimates	Effective stress-energy, modified field equation, selection/calibration revision	Role not invariant
Weak/strong lensing	Projected mass residual stabilizes deflection map	Geometric law modification, lensing-potential modification, source-plane/calibration changes	Role not invariant
CMB and large-scale structure	Cold component stabilizes acoustic peaks and growth history	Modified perturbation dynamics, extra fields, initial/boundary ledger, transport correction	Role not invariant

This table is not an argument that all alternatives are empirically equal. They are not. It is an argument that the ontological conclusion does not follow merely from the residual role. To extract dark matter as primitive structure one needs more than “whatever makes the metric surrogate close.” One needs an invariant extraction protocol for the object itself.

Proposition 2 (Particle-candidate separation). *The boundary result for dark matter as a sector does not rule out a particle candidate. If a particle is detected through a standing-preserving particle-physics protocol, the particle receives its standing from that protocol, not from the prior residual role predicate ρ_{DM}^S .*

Proof. The residual role predicate is fixed by surrogate closure. A direct particle detection would be fixed by a different witness protocol: production, scattering, decay, cross-section constraints, detector records, and identity-preserving replication across laboratories and descriptions. If such a protocol succeeds, it establishes standing for the detected object. It does not show that the earlier role predicate, defined as whatever closes the gravitational ledger, was already invariantly extracted as the same primitive entity-class. $\square$

Thus the correct boundary statement is precise: dark matter is admissible as a metric-surrogate stabilizer; dark matter does not achieve primitive ontological extraction from that stabilizer role alone.

8 Dark energy as residual acceleration role

Dark energy has the same structure with a different witness profile. In minimal Λ CDM, late-time acceleration is captured by Λ , or equivalently by an effective vacuum-energy component with equation of state $w = -1$. More general parameterizations introduce $w(a)$, scalar fields, effective fluids, or modified gravitational dynamics [8–10, 16, 17].

At the background level, once radiation, matter, curvature, and calibration choices are fixed, one can define a residual acceleration component by rearranging the Friedmann equation:

$$\Omega_{\text{DE}}^S(a) = \frac{H^2(a)}{H_0^2} - \Omega_r a^{-4} - \Omega_m a^{-3} - \Omega_k a^{-2}. \quad (16)$$

For Λ CDM this residual is constant. In a broader model family it may be dynamical, geometrized, absorbed into modified field equations, or represented by an effective source. Again, the empirical issue is not whether acceleration witnesses exist. They do. The classificatory issue is whether the residual's role identity is invariant under admissible redescription. It is not.

The cosmological constant makes the non-invariance especially visible. As a term in the Einstein equations, Λ may be written on the geometric side or as an effective stress-energy. As a parameter in the action, it may be treated as a coupling. As an effective cosmological fluid, it may be assigned pressure and density. As a phenomenological placeholder, it may mark failure of a restricted background model. These are not all the same extraction claim. The witness constraints do not by themselves select one classification as primitive.

Proposition 3 (No substance follows from acceleration). *Late-time acceleration licenses a residual acceleration term in a metric surrogate. It does not by itself extract a dark-energy substance.*

Proof. The acceleration witnesses constrain the expansion history and related observables. The assignment of the residual to a substance, a constant curvature term, a vacuum energy, a scalar field, a modified law, or an emergent variable is a decomposition choice unless an invariant witness protocol selects one of those classifications across admissible redescriptions. Without such a protocol, the residual has surrogate role but not invariant extraction as a substance. $\square$

This does not deny that Λ may be a real parameter of a successful theory. It denies that the dark-energy sector, defined as the residual that stabilizes acceleration in a metric surrogate, is thereby an invariantly extracted primitive constituent.

9 Restricted redescription and load transfer

The strongest orthodox-facing formulation is not that all cosmological models are equally good, mutually equivalent, or interchangeable. Many alternatives to Λ CDM fail in some regimes, and even successful alternatives usually preserve only a restricted subset of the full cosmological witness structure. The claim is narrower: within a fixed witness domain and declared comparison target, dark-sector primitive extraction cannot be inferred from a decomposition coordinate when admissible alternatives can redistribute the relevant stabilizing burden without loss of that domain's witness content.

Let a cosmological realization certificate be represented schematically as

$$\text{Cert}_C = (\text{Anc}_C, \text{Ten}_C, Q_C, \text{T}_C, \text{R}_C, \text{I}_C, \text{E}_C), \quad (17)$$

where anchors fix the domain, tensor witnesses fix standing-bearing content, Q_C fixes identity through the quotient, T_C fixes admissible continuation or transport between regimes, R_C fixes records and boundary conditions, I_C fixes estimators, and E_C fixes the error discipline. This is the AASC import used here: the witness content must factor through a stable realization certificate rather than through arbitrary presentation choices [24, 26, 27].

A dark-sector term may carry load in any of the following coordinates:

1. an added stress-energy component;
2. a constant curvature term;

3. a scalar, vector, tensor, or auxiliary field;
4. a modified gravitational or inertial law;
5. an initial-condition or boundary-record correction;
6. a calibration, selection, or likelihood correction;
7. an emergent or coarse-grained variable.

A successful model must of course satisfy observations. But if the fixed witness constraints relevant to a given inference are preserved while the load moves among these coordinates, the load-bearing coordinate is not itself the standing-bearing object. It is part of the skin of the surrogate unless independently fixed as tensor content. The argument requires only such restricted load transfer; it does not require exact global equivalence among Λ CDM, MOND-like theories, scalar-tensor models, emergent-gravity programs, or effective-fluid parameterizations.

Lemma 2 (Load-transfer lemma). *For a fixed witness set $\mathcal{W}$ and declared comparison target, any term whose identity is exhausted by the location of compensating load inside Cert_C fails standing unless the load location is invariant under the admissible redescription preserving that fixed witness content.*

Proof. If the term’s identity is the load location, and the load location changes while the fixed witness set and comparison target are preserved, then the term’s identity changes without a corresponding witness-level change in that domain. Standing requires identity preservation under admissible redescription. Therefore the term fails standing unless the load location itself is invariant. $\square$

This is why the presently inferred dark-sector structure is vulnerable in a way ordinary detected entities are not. Its sector-level definition is load-location language: missing mass, missing acceleration, additional gravitating contribution, unknown component, effective fluid. Those descriptions are scientifically useful, but the ontological failure is not merely ordinary underdetermination. It is a failure to produce an invariant extraction of the same entity-class from the fixed witness content.

10 Why empirical success does not close the boundary

A common objection is that Λ CDM works too well for its dark-sector terms to be merely surrogate-dependent stabilization structures. The objection conflates three claims:

1. the surrogate is empirically successful;
2. the dark-sector coordinates are indispensable to that surrogate;
3. the dark-sector coordinates achieve primitive ontological extraction.

The first two may be true while the third fails. A gauge choice, a coordinate chart, a renormalization scale, an effective potential, or a phenomenological parameter can be indispensable to a computation without being an invariantly extracted primitive constituent. The analogy is imperfect, as all analogies are, but the logical point is exact: model-internal necessity does not by itself establish standing.

Remark 2 (No anti-realism follows). *The argument is not global anti-realism about unobservables, nor anti-realism about cosmological ontology generally. It is a local boundary result about role-defined residuals. Many unobservables achieve invariant extraction because they are fixed by stable cross-witness protocols. The presently inferred dark-sector structure, as such, is not fixed in that way.*

Remark 3 (No denial of anomalies). *The argument also does not deny anomalies. Rotation curves, cluster lensing, CMB peaks, structure growth, and acceleration remain exactly as constraining as before. The theorem says that the residual required to compress them in a metric surrogate is not automatically invariantly extracted as a primitive entity-class.*

The same point applies to Bayesian confirmation. If a model with dark-sector parameters has high posterior support relative to a comparison class, that supports the model within that comparison class. It does not show that the parameters possess invariant identity across admissible redescriptions, especially when the comparison class itself fixes the metric-source decomposition in which those parameters live.

This also separates the present claim from ordinary underdetermination. The paper does not rest on the weak claim that several theories can fit the same data. It rests on the stronger extraction claim that the dark-sector role term itself is not invariantly recoverable from the fixed witness set once decomposition choices are allowed to vary within the admissible scope. The missing element is not certainty about which theory is true; it is an identity-preserving reconstruction map for the alleged entity-class.

11 Objection map

The following objections are common enough to warrant explicit treatment.

Objection	Boundary response
“Dark matter is observed gravitationally.”	The witnesses are gravitational. The question is whether the inferred residual is invariantly classified as matter rather than as modified law, auxiliary field, boundary correction, or effective source. Gravitational evidence supports a stabilizing contribution; it does not by itself supply sector-level primitive extraction.
“The Bullet Cluster and lensing separation prove dark matter.”	They strongly constrain many modified-gravity and baryon-only accounts and support a nonbaryonic gravitating component inside GR-like modeling. The boundary remains: the sector term achieves primitive extraction only if its identity is standing-preserving, not merely because a class of alternatives is disfavored.
“ Λ CDM fits CMB and large-scale structure.”	Correct. The theorem accepts this. The fit establishes empirical adequacy of the surrogate and indispensability of its coordinates, not primitive extraction of every coordinate.

“Modified gravity has problems; therefore dark matter is real.”	Failure of some redescrptions narrows the model class. It does not convert a residual role definition into invariant extraction. Primitive extraction still requires an extraction protocol.
“Dark energy is just Λ , and Λ is in the field equations.”	A parameter in field equations can be real as a parameter without being a substance or primitive sector. Moving Λ between geometry and effective stress-energy already shows that the substance classification is not fixed by the equation alone.
“This is merely semantics.”	The distinction is operational: what remains invariant under witness-preserving redescription? If the answer changes with decomposition, the difference is not semantic; it is a failed criterion for primitive ontological extraction.
“This is just underdetermination.”	Ordinary underdetermination says that rival theories may share evidence. The boundary claim is stronger: the dark-sector role term lacks a standing-preserving extraction map even before choosing among complete theories. The problem is invariant recovery of the entity-class, not merely lack of final theory choice.
“Future detection could vindicate dark matter.”	Yes for the detected object, if the detection protocol is standing-preserving. No for the prior role predicate as such. The residual label and the detected entity would need to be connected by an additional bridge theorem.

12 Positive program: what would count as standing?

The boundary theorem is negative only toward illicit promotion. It gives a positive test for future work. A dark-sector candidate gains standing only if it satisfies all of the following:

1. **Independent extraction:** the object is identified by a witness protocol not reducible to “whatever closes the metric surrogate.”
2. **Redescription invariance:** the object’s identity persists under admissible changes of coordinates, gauges, source-geometry splits, estimator choices, and observationally witness-preserving model presentations.
3. **Cross-witness coherence:** independent witness families converge on the same identity rather than merely on the same residual role.
4. **Bridge closure:** if the object is claimed to be the dark-sector residual, a bridge theorem connects the independently extracted object to the cosmological role without circularly defining it by that role.

For dark matter, this would resemble a particle or field detection that coherently bridges laboratory, astrophysical, and cosmological witnesses. For dark energy, it would require more than

an expansion residual; it would require invariant identification of the responsible structure across geometry/source redescrptions and background/perturbation regimes.

This is a high bar, but it is not hostile to science. It is exactly the bar needed to distinguish ontology from compression.

13 Boundary closure

We can now state the closure result in the language of the original target.

Theorem 2 (Dark-sector boundary closure). *The presently inferred dark-sector structure, as encoded by sector terms introduced to stabilize metric cosmological surrogates, is a role-defined residual structure. It preserves predictive adequacy inside successful representations but fails standing-preserving extraction under the admissible redescrptions analyzed here. Therefore no presently available dark-sector construction achieves primitive ontological extraction solely from the surrogate structures treated in this boundary analysis.*

Proof. The presently inferred dark-sector structure satisfies the hypotheses of the classification-error theorem in the fixed scope. Its dark-matter and dark-energy sector terms are introduced as additional stabilizing terms inside a metric surrogate. Each admits restricted witness-preserving load transfer, over the observational regimes motivating the corresponding inference, across at least some admissible changes of representation: source versus law, geometry versus effective stress-energy, auxiliary field versus effective fluid, boundary record versus parameter correction, or calibration/inference ledger versus physical component. Neither sector term, as a sector term, is extracted by a standing-preserving witness protocol independent of the surrogate role. Therefore the presently inferred structure fails standing-preserving extraction in the analyzed scope. Primitive ontological extraction is not achieved by current metric-surrogate indispensability alone. $\square$

The final boundary is therefore:

No presently available dark-sector construction achieves standing-preserving ontological extraction solely from the surrogate structures analyzed here. The presently inferred dark-sector structure may be predictively necessary inside metric-based compression without thereby achieving primitive ontological extraction.

No empirical result is erased by this closure. The observational constraints remain. Λ CDM remains a powerful compression. Dark-sector model building remains a legitimate scientific activity. Future detections or bridge theorems may introduce new standing-bearing objects for separate assessment. What is blocked here is only the illicit promotion from presently inferred role-defined surrogate stabilizer to standing-preservingly extracted constituent.

A AASC import ledger

This paper uses only the following AASC machinery [20–27].

1. **Standing invariance:** standing requires representation-independent reconstructability, i.e. identity preservation under admissible redescription.

2. **AMetric boundary discipline:** pre-admissible presentation structure does not generate primitive ontological extraction by metric, ranking, or selector alone.
3. **Anchor/tensor/skin decomposition:** anchors fix admissibility preconditions, tensors carry standing-bearing invariant content, and skin is representational surplus with no authority to generate primitive ontological extraction.
4. **Fixed-domain exhaustion:** within a fixed scope, an alleged same-domain discriminator either factors through existing invariant content, changes scope, or collapses to presentation-dependent stabilization structure.
5. **Cosmological transport discipline:** early/late or cross-regime cosmological claims require a stable realization certificate preserving anchors, tensor witnesses, quotient identity, records, estimators, and error discipline.

No stronger AASC thesis is needed. In particular, the paper does not rely on a new empirical prediction, a replacement metric theory, or a denial of standard cosmological data.

B Minimal formal normal form

Let $\mathcal{W}$ be the fixed witness set and let

$$\mathcal{S} = (\mathcal{M}, g, \Theta, \mathbf{P}, \mathbf{I}, \mathbf{R}, \mathbf{E}) \quad (18)$$

be a metric surrogate. Let D denote a proposed dark-sector coordinate. Define the residual map

$$\mathcal{D}_{\mathcal{S}}(\mathcal{W}) = \mathbf{I}_{\mathcal{S}}(\mathcal{W})|_D. \quad (19)$$

This map extracts the dark-sector coordinate assigned by $\mathcal{S}$. It is standing-preserving only if for every admissible redescription $R : \mathcal{S} \rightarrow \mathcal{S}'$ preserving the fixed witness set and comparison target, there exists an identity-preserving comparison isomorphism ι_R such that

$$\iota_R(\mathcal{D}_{\mathcal{S}}(\mathcal{W})) = \mathcal{D}_{\mathcal{S}'}(\mathcal{W}). \quad (20)$$

If no such family $\{\iota_R\}$ exists, the dark-sector coordinate is presentation-dependent. If the coordinate disappears under R , or is reclassified as geometry, modified law, boundary condition, or effective field without witness loss, then ι_R is unavailable unless one imports a new discriminator. Importing that discriminator changes the scope; failing to import it leaves the term without standing.

This yields the normal form:

$$\text{dark-sector inference} = \text{witness constraint} + \text{metric surrogate} + \text{residual load assignment}. \quad (21)$$

Only the first item is directly fixed by observation. The second is a licensed representation. The third is a surrogate-dependent stabilization assignment unless separately stabilized by invariant extraction.

C Compact referee checklist

A referee wishing to reject the boundary result must identify which premise fails:

1. The presently inferred dark-sector structure is not role-defined residual structure in metric surrogate inference.
2. Metric-source decompositions are invariant under all admissible redescrptions.
3. Predictive indispensability entails primitive ontological extraction.
4. A standing-preserving witness protocol already extracts the dark sector as a sector, not merely candidate objects or residual parameters.
5. Same-witness load transfer is impossible even locally or formally.

If none of these is established, the boundary stands. The presently inferred dark-sector structure may remain indispensable to a model while failing invariant ontological extraction.

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